

P-14: AI Lie Detector — Detecting AI-Generated Text

Team Members

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1. Problem Statement

As large language models (LLMs) such as ChatGPT and Gemini become more advanced, distinguishing between human-written and AI-generated text has become increasingly challenging. This poses concerns in areas such as academic integrity, misinformation, and authorship verification.

The goal of this project is to develop an “AI Lie Detector” — a system that can analyze text and determine whether it was generated by an AI model or written by a human. We aim to explore linguistic, statistical, and semantic indicators that reveal differences in writing styles, using explainable and lightweight methods rather than deep, opaque models.

2. Related Work

Several studies have examined methods for detecting AI-generated content.

- Perplexity-based approaches (Gehrmann et al., 2019) assess how predictable a piece of text is to a language model, with AI-generated text often having lower perplexity.
- Feature-based detection (Ippolito et al., 2020) uses linguistic features such as sentence length, word frequency, and part-of-speech distributions.
- Neural classifier methods (Mitchell et al., 2023) employ fine-tuned transformers for binary classification between human and AI text, though they often lack interpretability.

Our project builds upon these ideas by focusing on interpretable, feature-based detection to analyze why AI text differs from human text, not just whether it does.

3. Possible Approaches

Our project will follow a systematic four-step approach to develop an AI text detection system. First, we'll collect a balanced dataset by gathering human-written text from reliable public sources like Reddit, Wikipedia Talk pages, and news comments, then generate corresponding AI text using

OpenAI's GPT models or open-source LLMs such as LLaMA and Mistral on identical prompts to create directly comparable pairs. Second, we'll perform feature engineering by extracting interpretable linguistic characteristics including perplexity (typically lower in AI text), lexical diversity and word frequency patterns, sentence length variance (more varied in human writing), and punctuation and syntactic structures, following methodologies outlined in resources like the ActiveState guide and Hugging Face implementations. Third, we'll train and evaluate lightweight machine learning models such as Logistic Regression, Random Forest, or SVM on these engineered features, using standard metrics including accuracy, precision, recall, and F1-score on balanced datasets, while optionally testing model transferability across different AI generators and text domains. Finally, we'll implement explainability and visualization components by creating visual representations of discriminative features like perplexity versus sentence variance and generating an "AI-likeness" confidence score that helps users understand the detection decision, drawing on interpretable scoring approaches used in tools like GLTR and other API-backed solutions to make the classification process transparent and accessible.

4. Project Plan

Week	Task	Deliverables
Week 8	Literature review and dataset design	Summary of existing techniques and finalized dataset sources
Week 9	Data collection and cleaning	Dataset of human and AI-generated texts
Week 10	Feature engineering and initial experiments	Scripts for feature extraction
Week 11	Model training and testing	Preliminary results and model comparison
Week 12	Visualization, report and presentation	Figures, analysis, and final report draft, presentation slides

5. Expected Outcomes

We expect to produce a lightweight and interpretable tool capable of identifying AI-generated text with reasonable accuracy. More importantly, the project will provide insights into the linguistic features that distinguish human writing from AI output.

References

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